

Heat, Mass and Momentum Transfer Analogy

Introduction to Heat and Mass Transfer
Incropera and Dewitt

CONTINUITY EQ^N & x-momentum EQ^N (B.L.)

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \underbrace{\frac{\partial^2 u}{\partial y^2}}_{\text{diffusion}} \quad (2) \quad \frac{\mu}{\rho}$$

ENERGY EQ^N

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \underbrace{\alpha}_{\text{diffusion}} \frac{\partial^2 T}{\partial y^2} + \cancel{\dot{q}_v} + \cancel{\mu \Phi} \quad (iii) \quad \frac{\mu}{\rho C_p}$$

CONC. BL

$$u \frac{\partial C_A}{\partial x} + v \frac{\partial C_A}{\partial y} = \underbrace{D_{AB}}_{\text{diffusion}} \frac{\partial^2 C_A}{\partial y^2} + \cancel{\dot{r}_A} \quad (iv)$$

ADVECTION

DIFFUSION

THE CONVECTION TRANSFER EQUATIONS

2-D, STEADY FLOW

$$\frac{v_x}{U} \quad \frac{T_s - T}{T_s - T_\infty}, \quad \frac{C_{As} - C_A}{C_{As} - C_{A\infty}}$$

APPROXIMATIONS & SPECIAL CONSIDERATIONS :

INCOMPRESSIBLE, CONST. PROPERTIES, NEGLIGIBLE BODY FORCES, NON-REACTING ($\dot{N}_A = 0$); NO ENERGY GEN. ($\dot{Q}_V = 0$)
NEGLIGIBLE VISCOUS DISSIP. ($\phi = 0$).

B.L. APPROXIMATIONS

$$\left. \begin{array}{l} u \gg v \\ \frac{\partial u}{\partial y} \gg \frac{\partial u}{\partial x}, \quad \frac{\partial v}{\partial y}, \frac{\partial v}{\partial x} \end{array} \right] \text{VEL. B.L.}$$
$$\left. \begin{array}{l} \frac{\partial T}{\partial y} \gg \frac{\partial T}{\partial x} \end{array} \right] \text{THERMAL B.L.}$$
$$\left. \begin{array}{l} \frac{\partial C_A}{\partial y} \gg \frac{\partial C_A}{\partial x} \end{array} \right] \text{CONC. B.L.}$$

BOUNDARY LAYER SIMILARITY

- THE NORMALIZED CONVECTION TRANSFER EQ^Ns

NON DIMENSIONALIZING PARAMETERS

$$x^* = x/L, \quad y^* = y/L, \quad u^* = \frac{u}{V}, \quad v^* = \frac{v}{V}, \quad T^* = \frac{T - T_s}{T_\infty - T_s}, \quad C_A^* = \frac{C_A - C_{As}}{C_{A\infty} - C_A}$$

CONSERVATION EQNS

$$u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = -\frac{dp^*}{dx^*} + \underbrace{\frac{\gamma}{\nu L}}_{\text{circled}} \frac{\partial^2 u^*}{\partial y^{*2}}$$

$$u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} = \underbrace{\frac{\alpha}{\nu L}}_{\text{circled}} \frac{\partial^2 T^*}{\partial y^{*2}}$$

$\frac{1}{Re Pr}$

$$u^* \frac{\partial C_A^*}{\partial x^*} + v^* \frac{\partial C_A^*}{\partial y^*} = \underbrace{\frac{D_{AB}}{\nu L}}_{\text{circled}} \frac{\partial^2 C_A^*}{\partial y^{*2}}$$

$\frac{1}{Re Sc}$

BCs

WALL

$$u^*(x^*, 0) = 0$$

$$v^*(x^*, 0) = 0$$

$$T^*(x^*, 0) = 0$$

$$T^* \equiv \frac{T_s - T}{T_s - T_w}$$

$$C_A^*(x^*, 0) = 0$$

FREE STREAM

$$u^*(x^*, \infty) = \frac{u_{\infty}(x^*)}{V}$$

$$T^*(x^*, \infty) = 1$$

$$C_A^*(x^*, \infty) = 1$$

SIMILARITY PARAMETER

$$Re_L$$

$$Re_L, Pr$$

$$Re_L, Sc$$

FUNCTIONAL FORM OF THE SOLNS.

$$u^* = f_1(x^*, y^*, Re_L, \frac{dp^*}{dx^*})$$

$$x^* = x/L, y^* = y/L$$

$$u^* = u/V, v^* = v/V$$

SHEAR STRESS AT THE SURF. ($y^* = 0$)

$$\tau_s = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \frac{\mu V}{L} \left. \frac{\partial u^*}{\partial y^*} \right|_{y^*=0}$$

$$C_f = \frac{\tau_s}{\frac{1}{2} \rho V^2} = \frac{2}{Re_L} \left. \frac{\partial u^*}{\partial y^*} \right|_{y^*=0}$$

$$C_f = \frac{2}{Re_L} f_2(x^*, Re_L)$$

$$C_f \frac{Re_L}{2} = f_2(x^*, Re_L)$$

FOR A PRESCRIBED
~~GEOM~~ GEOMETRY

P_1

P_2

$P \neq P(y)$



$$\frac{P_2 - P_1}{\Delta x}$$

$$T^* = f_3(x^*, y^*, Re, Pr, \frac{dy^*}{dx^*})$$

CONV. H.T. COEFF.

$$h(T_s - T_\infty) = -k_f \frac{\partial T}{\partial y} \Big|_{y=0} \Rightarrow h = \frac{-k_f \frac{\partial T}{\partial y} \Big|_{y=0}}{T_s - T_\infty}$$

$$h = - \frac{k_f}{L} \frac{(T_\infty - T_s)}{(T_s - T_\infty)} \frac{\partial T^*}{\partial y^*} \Big|_{y^*=0}$$

$$Nu = \frac{hL}{k_f} = \frac{\partial T^*}{\partial y^*} \Big|_{y^*=0}$$

$$Nu = f_4(x^*, Re, Pr)$$

FOR A PRESCRIBED
GEOMETRY

AV. Nu
NUMBER

$$\overline{Nu} = \frac{\overline{h}L}{k_f} = f_5(Re, Pr)$$

$$C_A^* = f_6(x^*, y^*, Re_L, Sc, \frac{dp^*}{dx^*})$$

$$N_A'' = -D_{AB} \frac{\partial C_A}{\partial y} \Big|_{y=0} = \underline{\underline{h_m}} (C_{As} - C_{A\infty}) \quad \text{CONV. MT COEFF.}$$

$$h_m = \frac{-D_{AB} \partial C_A / \partial y \Big|_{y=0}}{C_{As} - C_{A\infty}}$$

$$h_m = \frac{D_{AB}}{L} \frac{\partial C_A^*}{\partial y^*} \Big|_{y^*=0}$$

$$\frac{h_m L}{D_{AB}} = \text{SHERWOOD NO.} = \frac{\partial C_A^*}{\partial y^*} \Big|_{y^*=0}$$

$$Sh = f_7(x^*, Re_L, Sc)$$

PRESCRIBED GEO.

$$\overline{Sh} = \frac{\overline{h_m} L}{D_{AB}} = f_8(\underline{Re_L}, Sc)$$

CONSERVATION EQNS

$$u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = -\frac{dp^*}{dx^*} + \underbrace{\frac{\gamma}{\nu L}}_{\text{circled}} \frac{\partial^2 u^*}{\partial y^{*2}}$$

$$u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} = \underbrace{\frac{\alpha}{\nu L}}_{\text{circled}} \frac{\partial^2 T^*}{\partial y^{*2}}$$

$\frac{1}{Re Pr}$

$$u^* \frac{\partial C_A^*}{\partial x^*} + v^* \frac{\partial C_A^*}{\partial y^*} = \underbrace{\frac{D_{AB}}{\nu L}}_{\text{circled}} \frac{\partial^2 C_A^*}{\partial y^{*2}}$$

$\frac{1}{Re Sc}$

BCs

WALL

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FREE STREAM

$$u^*(x^*, \infty) = \frac{u_{\infty}(x^*)}{V}$$

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SIMILARITY
PARAMETER

Re_L

Re_L, Pr

Re_L, Sc

THE REYNOLD'S ANALOGY

$$\frac{dp^*}{dx^*} = 0 \quad \underline{Pr = Sc = 1} \Rightarrow \text{@ SOLNS OF } U^*, T^*, C_A^* \text{ MUST BE EQUIVALENT}$$

$$f_1 = f_3 = f_6$$

Geometric Similarity

SAME IS TRUE FOR C_f , Nu , Sh .

Dynamic Similarity

$$f_2 = f_4 = f_7$$

$$C_f \frac{Re_L}{2} = Nu = Sh.$$

$$Pr = Sc = 1$$

$$\frac{C_f}{2} = \frac{Nu}{Re_L (Pr)} = \frac{Sh}{Re_L (Sc)}$$

$$\frac{Nu}{Re Pr} \equiv \text{STANTON No. for HT}$$
$$\frac{Sh}{Re Sc} \equiv \text{STANTON No. for MASS TR.}$$

$$\frac{C_f}{2} = St = St_m$$

- REYNOLD'S ANALOGY

$$u^* = f_1(x^*, y^*, Re_L, \frac{dp^*}{dx^*})$$

$$T^* = f_3(x^*, y^*, Re_L, Pr, \frac{dp^*}{dx^*})$$

$$C_A^* = f_6(x^*, y^*, Re_L, Sc, \frac{dp^*}{dx^*})$$

$$C_f \frac{Re_L}{2} = f_2(x^*, Re_L)$$

$$Nu = f_4(x^*, Re, Pr)$$

FOR A PRESCRIBED
GEOMETRY

$$Sh = f_7(x^*, Re, Sc)$$

PRESCRIBED GEO.

MODIFIED REYNOLD'S ANALOGY

CHILTON ~~AND~~ COLBURN ANALOGY

$$\frac{C_f}{2} = St \cdot \underline{Pr}^{2/3} = j_H$$

$$0.6 < Pr < 60$$

$$\frac{C_f}{2} = St_m \cdot \underline{Sc}^{2/3} = j_m$$

$$0.6 < Sc < 300$$

$$\frac{C_f}{2} = St \cdot Pr^{2/3} = St_m \cdot Sc^{2/3}$$

$$\boxed{\frac{C_f}{2} = j_H = j_m}$$

$j \equiv$ COLBURN ' j ' FACTOR

BOUNDARY LAYER ANALOGIES

$$Pr = \frac{\nu}{\alpha} = \frac{\text{MOMENTUM DIFFUSIVITY}}{\text{THERMAL "}}$$

FOR LAMINAR BL

$$\frac{\delta}{\delta_t} = Pr^n, \quad \text{FOR A GAS } \underline{\delta_t \propto \delta},$$

FOR LIQ. METAL $\delta_t \gg \delta$, FOR OIL

$$Sc = \frac{\nu}{D_{AB}} = \frac{M^2 \text{ DIFF.}}{\text{MASS "}}, \quad \frac{\delta}{\delta_c} = Sc^n.$$

LEWIS NO. Le = $\frac{\text{THERMAL DIFF.}}{\text{MASS DIFF.}} = \frac{\alpha}{D_{AB}} = \frac{\nu}{D_{AB}} \cdot \frac{\alpha}{\nu} = \frac{Pr}{Sc}$

FOR SIMULTANEOUS HEAT & MASS TR

$$m = 1$$

INCRP
DEWIT
INTRO
HEAT&

Chilton Colburn Analogy (Modified Reynold's Analogy)

$$\frac{C_f}{2} = St \cdot Pr^{2/3} = j_h = St_m \cdot Sc^{2/3} = j_m \quad \begin{matrix} 0.6 < Pr < 60 \\ 0.6 < Sc < 300 \end{matrix}$$

	Momentum	Heat	Mass
Laminar	$C_{f,x} = 0.664 Re_x^{-1/2}$ $Re_x < 5 \times 10^5$	$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$ $\overline{Nu}_L = 0.664 Re_L^{1/2} Pr^{1/3}$ $Re_x < 5 \times 10^5, 0.6 < Pr < 50$	$Sh_x = 0.332 Re_x^{1/2} Sc^{1/3}$ $\overline{Sh}_L = 0.664 Re_L^{1/2} Sc^{1/3}$ $Re_x < 5 \times 10^5, 0.6 < Sc < 300$
Mixed/ Turbulent	$C_{f,L} = \frac{0.074}{Re_L^{1/5}} - \frac{1742}{Re_L}$ $5 \times 10^5 < Re_L < 10^8$ $Re_{x,c} = 5 \times 10^5$	$\overline{Nu}_L = (0.037 Re_L^{4/5} - 871) Pr^{1/3}$ $0.6 \leq Pr \leq 50, Re_{x,c} = 5 \times 10^5$ $5 \times 10^5 < Re_L < 10^8$	$\overline{Sh}_L = (0.037 Re_L^{4/5} - 871) Sc^{1/3}$ $0.6 \leq Sc \leq 300, Re_{x,c} = 5 \times 10^5$ $5 \times 10^5 < Re_L < 10^8$